

The idea is to build a small humanoid-like robot, roughly 17 inches tall, designed not as a miniature industrial worker but as a physical AI companion. Its purpose would be to provide intellectual assistance, continuity, memory, conversation, timekeeping, and companionship while remaining small, approachable, relatively safe, and inexpensive compared with full-sized humanoid robots.

The central idea is that a robot does not need to be human-sized to provide meaningful assistance. A large humanoid is useful when it needs to interact with a human-sized physical environment, such as carrying objects, opening doors, climbing stairs, or performing industrial labor. A small robot can instead concentrate on cognitive and social assistance. It can remember appointments, maintain notes and timelines, remind someone about tasks, record information when requested, communicate with family members, answer questions about recent events, and provide a persistent conversational presence throughout the day.

The robot would function as a kind of physical memory and timekeeping system. Rather than requiring a person to pick up a phone and interact with an application, the assistant would simply exist in the environment. It could notice when someone speaks to it, turn toward them, walk over, speak, gesture, and maintain continuity between interactions. For elderly people or people who struggle with memory, this continuity could potentially be especially valuable. The robot would not replace human caregivers, but could serve as an additional layer of everyday support.

The physical design would intentionally avoid trying to reproduce a perfectly human body. A small, creature-like design inspired by Mewtwo could use thick, wide legs and a relatively narrow torso to provide a low center of gravity and good stability. The robot would prioritize standing, walking, falling safely, and recovering from falls rather than attempting difficult activities such as climbing stairs. Its intended environment would primarily be indoor floors, carpets, rugs, and mild outdoor slopes or uneven terrain.

The underlying software architecture would be based on NatOS. NatOS would serve as the physical operating layer between the AI system and the robot's hardware. It would manage sensors, actuators, power, communication, timing, diagnostics, and other hardware interfaces while allowing a higher-level AI system to reason about the environment and interact with the user.

The architecture could eventually resemble a layered system in which the AI provides high-level intentions, NatOS translates those intentions into physical behaviors, and sensors continuously provide feedback about what actually happened. The AI might decide that the robot should walk toward a person, while NatOS handles the much less glamorous but essential work of controlling motors, maintaining balance, respecting mechanical limits, and responding to failures.

Development should proceed incrementally rather than attempting to build the entire humanoid immediately. The first practical milestone is a handheld Bluetooth-enabled device running NatOS. After that, individual robotic actuators and limbs can be developed and tested independently. A single joint can establish reliable motor control and feedback; a small multi-joint limb can establish kinematics; a leg can develop standing and balance; two legs can establish walking; arms can provide gestures and recovery behavior; and only then should the complete body be assembled.

This approach also allows NatOS to mature alongside the hardware. The same operating system that manages a handheld scanner can eventually manage motors, encoders, IMUs, cameras, microphones,

batteries, and other robotic components. The robot therefore becomes another physical platform for the operating system rather than an entirely separate project.

The long-term vision is not necessarily to build the most capable humanoid robot possible. It is to determine how much useful intelligence can be given a relatively small, affordable physical body. A full-sized humanoid may be necessary for physical labor, but a small embodied AI may be sufficient for memory, conversation, organization, companionship, and continuity of everyday life.

The ultimate product would therefore be less like a miniature industrial robot and more like a physical personal intelligence: a small creature that can see, hear, move, remember, speak, keep track of time, and participate in a person's daily life. The body provides presence and agency; the AI provides intelligence; memory provides continuity; and NatOS provides the interface between all of them and the physical world.